Class 6

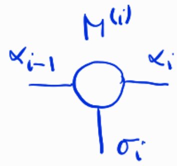
Reminder

Matrix Product States

$$|\psi\rangle = \sum_{\sigma_1 \dots \sigma_L} \underline{\underline{M}}^{(1)\sigma_1} \underline{\underline{M}}^{(2)\sigma_2} \dots \underline{\underline{M}}^{(L)\sigma_L} |\sigma_1 \dots \sigma_L\rangle$$

$\mathrm{size}(\underline{\underline{M}}^{(i)}) = M \times d \times M$

"bond" dimension



$\triangleright$ smaller sizes at chain ends:

$\dim(\alpha_i) \leq d^i$

$\dim(\alpha_i) \leq d^{L-i}$

special case

$\mathrm{size}(M^{(1)}) = 1 \times d \times d$

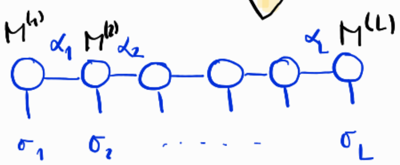
$\mathrm{size}(M^{(L)}) = d \times d \times 1$

Graphical representation

$\underline{\underline{M}}^{(1)\sigma_1} \underline{\underline{M}}^{(2)\sigma_2} \dots \underline{\underline{M}}^{(L)\sigma_L}$

$\sum_{\alpha_1 \dots \alpha_{L-1}} M^{(1)\sigma_1}_{\alpha_1} M^{(2)\sigma_2}_{\alpha_1\alpha_2} \dots M^{(L)\sigma_L}_{\alpha_{L-1}}$

$\Updownarrow$



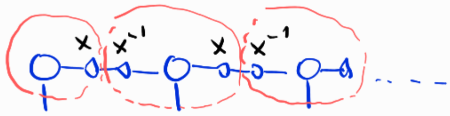
Connected leg

$\Updownarrow$

Summation

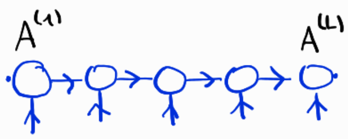
Gauge freedom

$$\underbrace{\underline{\underline{M}}^{(1)\sigma_1} \underline{\underline{X}}^{(1)}}_{\tilde{\underline{\underline{M}}}^{(1)\sigma_1}} \underbrace{\underline{\underline{X}}^{(1)^{-1}} \underline{\underline{M}}^{(2)\sigma_2} \underline{\underline{X}}^{(2)}}_{\tilde{\underline{\underline{M}}}^{(2)\sigma_2}} \underline{\underline{X}}^{(2)^{-1}} \dots \underbrace{\underline{\underline{X}}^{(L-2)^{-1}} \underline{\underline{M}}^{(L-1)\sigma_{L-1}} \underline{\underline{X}}^{(L-1)}}_{\tilde{\underline{\underline{M}}}^{(L-1)\sigma_{L-1}}} \underbrace{\underline{\underline{X}}^{(L-1)^{-1}} \underline{\underline{M}}^{(L)\sigma_L}}_{\tilde{\underline{\underline{M}}}^{(L)\sigma_L}}$$



Gauge fixing

left canonical: $\underline{\underline{A}}^{(1)\sigma_1} \underline{\underline{A}}^{(2)\sigma_2} \dots \underline{\underline{A}}^{(L)\sigma_L}$



$$\sum_{\alpha_{i-1}\sigma_i} \left(A^{(i)\sigma_i}_{\alpha_{i-1}\beta_i}\right)^* A^{(i)\sigma_i}_{\alpha_{i-1}\alpha_i} = \delta_{\beta_i\alpha_i}$$

$A^{(i)}$'s are (left) isometries

$A^+A = 1$

$AA^+ \neq 1$!

$\Downarrow$

projector on kept states

reshaped to

$A_{(\alpha_{i-1}\sigma_i)(\alpha_i)}$

$Md \times M$

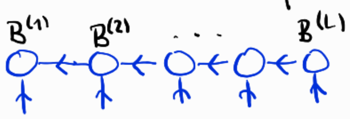
Right canonical $\underline{\underline{B}}^{(1)\sigma_1} \dots \underline{\underline{B}}^{(L)\sigma_L}$

$$\sum_{\alpha_i\sigma_i} \left(B^{(i)\sigma_i}_{\beta_{i-1}\alpha_i}\right)^* B^{(i)\sigma_i}_{\alpha_{i-1}\alpha_i} = \delta_{\beta_{i-1}\alpha_{i-1}}$$

$B^{(i)}$'s are right isometries

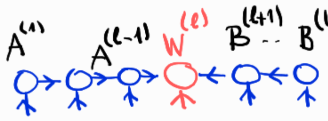
$BB^+ = 1$

$B^+B \neq 1$



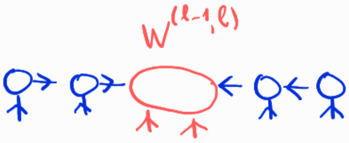
Mixed canonical

- with 1-site "core"



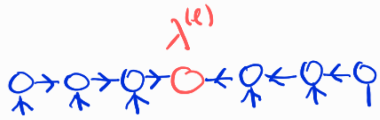
full wavefunc. $$|\psi\rangle = \sum_{\substack{\alpha_{l-1}\alpha_l \\ \sigma_l}} W^{(l)\sigma_l}_{\alpha_{l-1}\alpha_l} \underbrace{|\alpha_{l-1}\rangle_L \otimes |\sigma_l\rangle \otimes |\alpha_l\rangle_R}_{\text{This is an orthogonal basis of the (truncated) Hilbert space}}$$

- with 2-site "core"



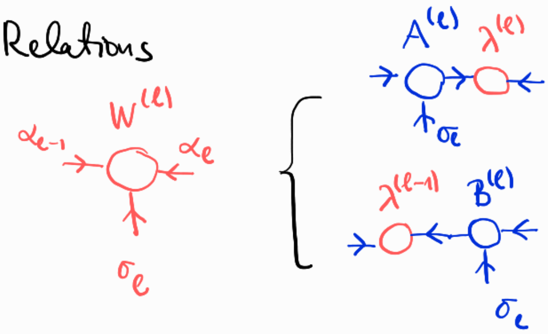
Usual form in DMRG iterations

- with 0-site "core"



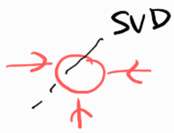
if W is diagonal with non-negative entries $\Updownarrow$ Schmidt decomp

Relations



- from $W^{(l)}$ we can get $A^{(l)}, B^{(l)}$ by SVD

$\Rightarrow A^{(l)}, \lambda^{(l)}, V$



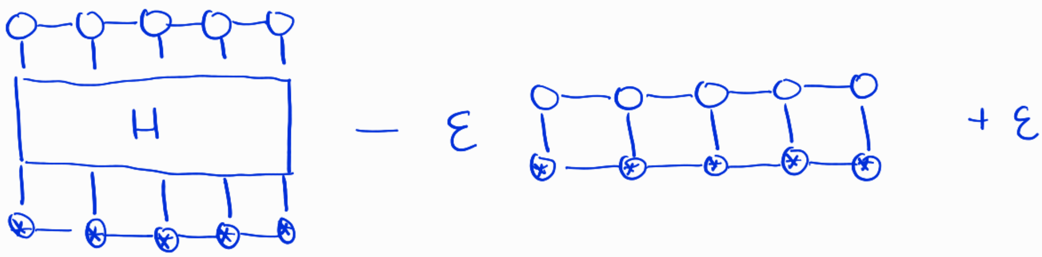
$\Rightarrow U, \lambda^{(l-1)}, B^{(l)}$

DMRG as a variational algorithm

- MPS is now an ansatz, $M^{(l)\sigma_l}_{\alpha_{l-1}\alpha_l}$ are variational parameters

Goal:
- minimize $\langle\psi|\hat{H}|\psi\rangle$
- constraint $\langle\psi|\psi\rangle - 1 = 0 \Rightarrow$ Lagrange multiplier ε

$$\langle\psi|H|\psi\rangle - \varepsilon(\langle\psi|\psi\rangle - 1) = \sum_{\sigma_1\dots\sigma_L}\sum_{\tilde\sigma_1\dots\tilde\sigma_L} \underline{\underline{M}}^{(1)\tilde\sigma_1 *} \dots \underline{\underline{M}}^{(L)\tilde\sigma_L *} H^{\tilde\sigma_1\dots\tilde\sigma_L}_{\sigma_1\dots\sigma_L} \underline{\underline{M}}^{(1)\sigma_1} \dots \underline{\underline{M}}^{(L)\sigma_L} - \varepsilon \sum_{\sigma_1\dots\sigma_L} \left(\underline{\underline{M}}^{(1)\sigma_1 *} \dots \underline{\underline{M}}^{(L)\sigma_L *}\right)\left(\underline{\underline{M}}^{(1)\sigma_1} \dots \underline{\underline{M}}^{(L)\sigma_L}\right) + \varepsilon$$



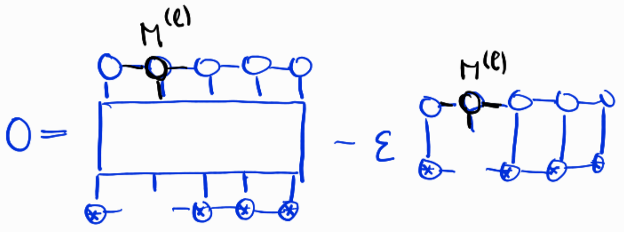
$\frac{\partial}{\partial M^{(\ell)*}}$

- independent minimization for M & M^*
- expression is linear in $M^{(\ell)*}$

$\Downarrow$

$\rightarrow$ $M^{(\ell)\sigma_\ell}_{\alpha_{\ell-1}\alpha_\ell}$ is removed

$\rightarrow$ No summation for σ_ℓ, $\alpha_{\ell-1}\,\alpha_\ell$



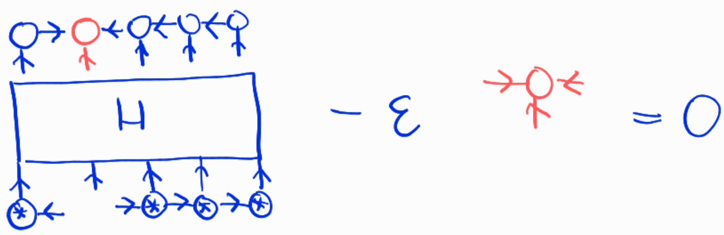
This is a generalized eigenvalue problem for $M^{(\ell)}$!

ε: this is the eigenvalue

Using mixed canonical MPS here makes life easier

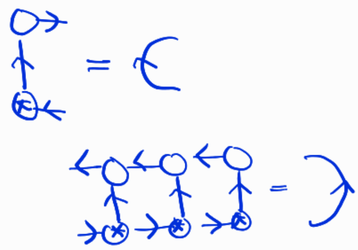
$\underline{\underline{A}}^{(1)\sigma_1} \cdots \underline{\underline{A}}^{(\ell-1)\sigma_{\ell-1}}\ \underline{\underline{M}}^{(\ell)\sigma_\ell}\ \underline{\underline{B}}^{(\ell+1)\sigma_{\ell+1}} \cdots \underline{\underline{B}}^{(L)\sigma_L}$

$\Rightarrow$



Standard Symmetric (Hermitean) eigen problem. $\Leftarrow$ Lánczos

$\Leftarrow$ because (isometry)



Note: good convention: conjugation flips arrow directions

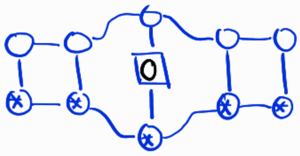
Important! Storage of H is a delicate issue ("Tables", MPO)

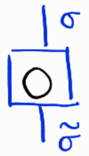
- This is 1-site DMRG
- Sweeping: $\rightarrow$ after solving $M^{(\ell)}$ we move to $\ell \pm 1$ (left / right sweep)
  $\rightarrow$ moving the "core" position: SVD of $M^{(\ell)}$ $\Leftrightarrow$ reduced density matrix diagonalization
- 1 site DMRG is prone to getting stuck, and it is hard to increase bond dim.
  $\hookrightarrow$ 2-site DMRG: $\frac{\partial}{\partial (M^{(\ell-1)*} M^{(\ell)*})} = \emptyset$

Calculating observables / correlators

- Local observable $\hat{O}^{(\ell)} = \mathbb{1} \otimes \mathbb{1} \otimes \dots \otimes \hat{O} \otimes \mathbb{1} \otimes \mathbb{1}$

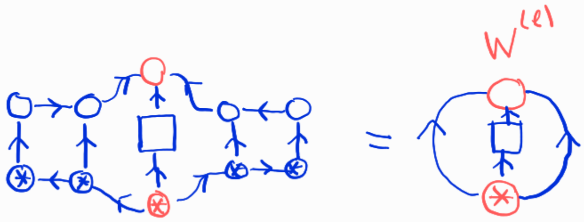
examples: $\hat{S}^z_\ell$, $\hat{S}^x_\ell$, $\hat{n}_\ell$

$\langle\psi|\hat{O}^{(\ell)}|\psi\rangle \Longleftrightarrow$ 

here  $= O_{\sigma\tilde{\sigma}}$

is the $d \times d$ (1-site) operator.

$\hookrightarrow$ mixed canonical case

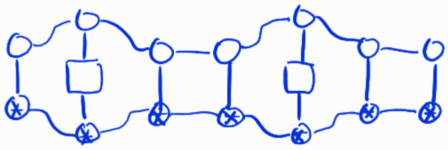
$\underline{A}^{(1)} \cdots \underline{A}^{(\ell-1)}\, \underline{W}^{(\ell)}\, \underline{B}^{(\ell+1)} \cdots \underline{B}^{(L)}$: 

- this is no surprise: $W^{(\ell)}$ are the wavefunction amplitudes in the orthogonal basis

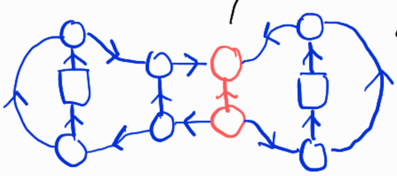
$|\alpha_{\ell-1}\rangle_L |\sigma_\ell\rangle |\alpha_\ell\rangle_R$

Correlators

$\langle O^{(\ell)} O^{(k)} \rangle$



$\Downarrow$ canonical MPS

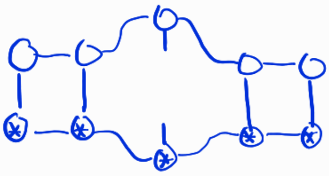


we can place the core anywhere in $\ell \leq i_{core} \leq k$

How to calculate general observables (many different O's ?)

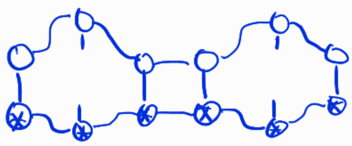
$\hookrightarrow$ 1-site reduced density matrix ?

$$\rho^{(\ell)} = \mathrm{Tr}_{all \setminus \ell}\left\{|\psi\rangle\langle\psi|\right\} \Longrightarrow$$



$\hookrightarrow$ 2-site RDM

$$\rho^{(\ell,k)} = \mathrm{Tr}_{all \setminus \ell, k}\left\{|\psi\rangle\langle\psi|\right\} \Longrightarrow$$



Note

for fermions modification is necessary (next class)

Entropies

$$S^{(\ell)} = -\mathrm{tr}\, \rho^{(\ell)} \log \rho^{(\ell)} \Longleftrightarrow$$

2-site mutual info

$$I^{(\ell,k)} = S^{(\ell)} + S^{(k)} - S^{(\ell,k)}$$

Symmetries & MPS

- Abelian symmetries

$\boxed{\text{Symmetry, if } [\hat{U}(g), \hat{H}] = \emptyset \quad \forall g \in \mathcal{G}}$

- Abelian group $\mathcal{G}$

$\Updownarrow$

$g_1 g_2 = g_2 g_1 \quad \forall g_1, g_2 \in \mathcal{G}$

- unitary representation (reducible)

$\hat{U}(g) \Leftrightarrow \hat{U}(g_1 g_2) = \hat{U}(g_1)\hat{U}(g_2) \quad \forall g_1 g_2 \in \mathcal{G}$

<u>Example</u> XXZ-model

$$H = J \sum_i \left(\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_{i+1}^z \right)$$

$$\hat{U}(\varphi) = e^{i2\varphi \sigma^z_{tot}}$$

$$\Rightarrow \quad [\sigma^z_{tot}, H] = \emptyset$$

- Locally generated global symmetry

$$e^{2i\varphi \sigma^z_{tot}} = e^{2i\varphi \sum_j \sigma_j^z} = e^{2i\varphi \sigma_1^z} e^{2i\varphi \sigma_2^z} \cdots e^{2i\varphi \sigma_L^z}$$

$$\hat{U}(g) = \hat{U}_{loc}(g) \otimes \hat{U}_{loc}(g) \otimes \cdots \otimes \hat{U}_{loc}(g)$$

$\Updownarrow$

Local (1-site) representation (dxd unitary)

Translational invariance is <u>not</u> such symmetry!

How to exploit this?

Schmidt decomposition

Symmetrical basis in A, B $\Leftarrow$ $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$

$$\hat{\sigma}^z_{tot} = \hat{\sigma}^z_A + \hat{\sigma}^z_B$$

$|i\rangle_A : \quad \hat{\sigma}^z_A |i\rangle_A = S^z_{A_i} |i\rangle_A$

$|j\rangle_B : \quad \hat{\sigma}^z_B |j\rangle_B = S^z_{B_j} |j\rangle_B$

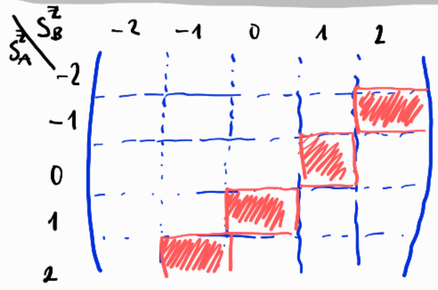
quantum number (eigenvalue)

$\boxed{|\psi\rangle : \quad \hat{\sigma}^z_{tot} |\psi\rangle = S^z |\psi\rangle}$

$$|\psi\rangle = \sum_{ij} \psi_{ij} \, |i\rangle \otimes |j\rangle$$

ψ_{ij}

$S^z = 1$
example



SVD: separately for every block (because only 1 block / row, 1 block / col is occupied)

$\Downarrow$

Schmidt decomposition

$$|\psi\rangle = \sum_\alpha \lambda_\alpha |\alpha, S^z_{A\alpha}\rangle_A \otimes |\alpha, \underbrace{S^z - S^z_{A\alpha}}_{S^z_{B\alpha}}\rangle_B$$

$S^z_{A\alpha}$, $S^z_{B\alpha}$ are good quantum numbers!

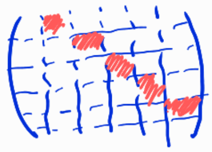
$|\alpha\rangle_A$, $|\alpha\rangle_B$ are eigenstates of $\hat{\sigma}^z_A$ & $\hat{\sigma}^z_B$

Consequences for MPS

$$\underset{S^z_{L\alpha_\ell}}{|\alpha_\ell\rangle_L} = \sum_{\sigma_\ell \alpha_{\ell-1}} \hat{A}^{(\ell)\sigma_\ell}_{\alpha_{\ell-1}\alpha_\ell} \underset{S^z_{L\alpha_{\ell-1}}}{|\alpha_{\ell-1}\rangle} \underset{s^z_\ell = \pm 1/2}{|\sigma_\ell\rangle}$$

$\underline{A}^{(\ell)\uparrow} =$ 

$S^{z(\ell)}_{L\alpha_\ell} = S^z_{L\alpha_{\ell-1}} + \frac{1}{2}$

$\underline{A}^{(\ell)\downarrow} =$ 

$S^{z(\ell)}_{L\alpha_\ell} = S^z_{L\alpha_{\ell-1}} - 1/2$

$$\boxed{S^z_{L\alpha_\ell} = S^z_{L\alpha_{\ell-1}} + s^z_\ell}$$

To exploit this

- Implement block-sparse tensors
- Block sparse tensor algebra, svd
- Block sparse operators

→ This is straight forward.

→ Form of MPS and algorithms like DMRG is practically unchanged.

Non-Abelian symmetries

Example: Heisenberg (xxx) chain

$$\hat{H} = J \sum_i \sigma^x_i \sigma^x_{i+1} + \sigma^y_i \sigma^y_{i+1} + \sigma^z_i \sigma^z_{i+1}$$

$$\hat{U}(\underline{n}, \varphi) = e^{i \underline{n} \cdot (\underline{\sigma}_{tot}) \cdot \varphi}$$

$|\psi; s, s^z\rangle \Longleftarrow$ S^2_{tot} & S^z_{tot} are good quantum numbers

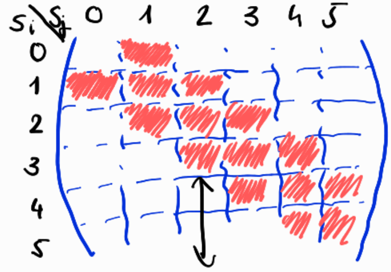
$$\sigma^2_{tot} |\psi; s, s^z\rangle = s(s+1)|\psi; s, s^z\rangle$$

$$\sigma^z_{tot} |\psi; s, s^z\rangle = s^z |\psi; s, s^z\rangle$$

Schmidt-decomposition?

$$|\psi\rangle = \sum_{i,j} \psi_{ij}\, |i, s_i\, s_i^z\rangle\, |j, s_j\, s_j^z\rangle$$

$\psi_{ij} =$



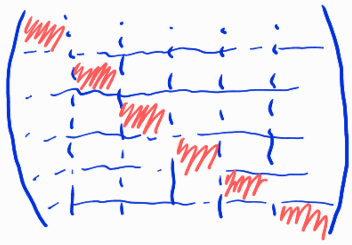
also extra block structure because of S^z.

ex: $S = 1$

- Not block diag!
- Schmidt states are not eigenstates of σ_i^2 & σ_j^2

Exception: $S = \emptyset$

$\psi_{ij} =$



$S_i = S_j$

$S_i^z = -S_j^z$

$\Downarrow$

$$|\psi\rangle = \sum_{\alpha} \lambda_\alpha\, |\alpha, S_\alpha, S^z_{\alpha A}\rangle |\alpha, S_\alpha, \underbrace{-S^z_{\alpha A}}_{S^z_{\alpha B}}\rangle$$

λ_α is degenerate within multiplets

α	τ	μ	S_A	S_A^z
1	1	1	0	0
2	2	1	1	−1
3	2	2	1	0
4	2	3	1	+1
5	3	1	0	0
6	4	1	2	−2
⋮				⋮

$\Leftarrow$ α → τ multiplet index; → μ internal index ($\sim s^z$)

SU(2) invariant MPS

General Non-Abelian groups

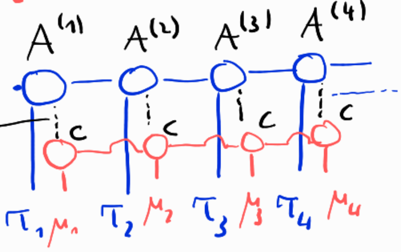
$|\psi\rangle$ must belong to the trivial representation

conjugate rep.

$$|\psi\rangle = \sum_{\Gamma} \sum_{\tau \in \Gamma} \Lambda_{\Gamma\tau} \sum_{\mu_1 \mu_2} C^{00}_{\Gamma\mu_1\, \bar{\Gamma}\mu_2}\, |\Gamma, \tau\mu_1\rangle_A\, |\bar{\Gamma}\, \tau\mu_2\rangle_B$$

Generalized Clebsch-Gordan coefficient

$$|s_e\, t_e\, m_e\rangle = \sum_{\substack{s_{e-1} t_{e-1} \\ s_e \tau_e}} A^{(e)\, s_e \tau_e}_{s_{e-1} t_{e-1}\, s_e t_e} \sum_{\substack{m_{e-1} \\ \mu_e}} C^{s_e m_e}_{s_{e-1} m_{e-1}\, s_e \mu_e}\, |s_{e-1}, t_{e-1}, m_{e-1}\rangle\, |s_e \tau_e \mu_e\rangle$$



dashed line: same s_{e-1} s_e s_e in A & C